



# SOWRAV CHOWDHURY

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## PERSONAL STATEMENT

I am an Electrical and Electronics Engineer specializing in embedded systems, microcontroller programming, and circuit design. My experience includes developing efficient, reliable solutions using platforms like Arduino and Raspberry Pi, as well as proficiency in CAD software and simulation tools. Skilled in programming for embedded electronics and optimizing control systems, I am committed to advancing innovative, high-performance solutions that meet real-world engineering challenges. My technical expertise in digital signal processing, power electronics, and circuit analysis equips me to contribute effectively to complex, multidisciplinary projects.

## EDUCATION

**Bachelor of Science in Electrical & Electronic Engineering** Oct 2018–Feb 2023

*University of Liberal Arts Bangladesh (ULAB)*

- Graduated with **Cum Laude**, CGPA: 3.77 (out of 4)
- Major: Electronics
- Thesis Project: Developed a portable, low-cost digital stethoscope using Raspberry Pi 4, achieving 81% accuracy in pneumonia detection through a 1D-CNN model and self-collected lung sound database.

**Cambridge International A'Levels** May 2016–Jun 2018

*St. Francis Xavier's Greenherald International School*

- CGPA: 2.5 (out of 5)

**Cambridge International O'Levels** May 2005–Jun 2015

*St. Francis Xavier's Greenherald International School*

- CGPA: 5 (out of 5)

## PUBLICATIONS

**Artificial Intelligence-Based Algorithms and Healthcare Applications of Respiratory Inductance Plethysmography: A Systematic Review**

*MDPI-Algorithms, Vol. 17, Issue 6, 2024, in a special issue on "Artificial Intelligence-based Algorithms with Potential Applications in Healthcare and Prediction of Disease Evolution."*

- This review examines Respiratory Inductance Plethysmography (RIP) as a non-invasive alternative to spirometry, highlighting that while RIP is promising for applications like apnea detection and human activity recognition via machine learning, it currently lacks the accuracy to replace spirometry in clinical use.
- DOI: <https://doi.org/10.3390/a17060261>

**Pediatric Pneumonia Diagnosis: Integration of a Self-Assembled Digital Stethoscope with Raspberry Pi and 1D CNN Model**

*2024 International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iACCESS), IEEE Xplore.*

- This study developed a low-cost digital stethoscope with a Raspberry Pi and 1D-CNN model, achieving 86% accuracy in detecting pneumonia in children aged 0-6. It offers a reliable, stand-alone diagnostic tool for remote areas lacking radiologic imaging.
- DOI: <https://doi.org/10.1109/iACCESS61735.2024.10499564>

**Quality Assessment of Respiratory Sounds Extracted from Self-Assembled Digital Stethoscopes**

*2022 International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE), IEEE Xplore.*

- This study developed and evaluated three low-cost digital stethoscopes for respiratory sound analysis, finding one with 84% feature correlation, which may support reliable data collection for machine learning models in diagnosing respiratory disorders.
- DOI: <https://doi.org/10.1109/ICAEEE54957.2022.9836502>

## WORK EXPERIENCES

### Research Assistant

Aug 2023–Present

*University of Liberal Arts Bangladesh (ULAB)*

- **Project:** This project aims to develop a low-cost digital stethoscope for automated pneumonia diagnosis in young children. It focuses on integrating advanced machine learning algorithms to improve diagnostic accuracy and accessibility in resource-limited settings.
- **Funded by:** ICT Innovation Fund, Ministry of Posts, Telecommunications and Information Technology, Government of Democratic Republic of Bangladesh.
- **Key Responsibilities:**
  - Developed a digital stethoscope from inception with a 3D-printed adapter to improve component integration.
  - Conducted hospital surveys and gathered respiratory sounds from pediatric patients to build a lung sound database.
  - Utilized advanced signal processing to enhance database quality.
  - Collaborated with healthcare professionals to ensure data integrity and refine diagnostic techniques.

### Teaching Assistant

Oct 2022–Sep 2023

*Department of Electrical and Electronic Engineering, ULAB*

- **Key Responsibilities:**
  - Assisted faculty with lab classes, course material preparation, and document management.
  - Graded lab reports, with feedbacks and improvements.
  - Provided student counseling and invigilation support.
  - Participated in faculty-led research projects.
- **Courses Assisted:**
  - Electrical Circuits I Lab (EEE 1102)
  - Microprocessor and Embedded Systems Lab (EEE 312)
  - Control Systems I Lab (EEE 404)
  - Digital Electronics Lab (EEE 2302)
  - VLSI I Lab (EEE 4425)

## ACHEIVEMENTS & AWARDS

### Awardee, Vice-Chancellor's Honors List

May 2021

*University of Liberal Arts Bangladesh (ULAB)*

Awarded for exceptional academic achievement and outstanding contribution to university co-curricular activities.

### Champion, Hult Prize OnCampus

May 2023

**Project:** *Autonomous Recycling of Waste Fabric using Augmented Reality & Machine Learning*

As Captain, led team to victory, demonstrating groundbreaking applications of AR and ML to promote sustainable waste management in garments industry.

### Semi-Finalist, Hult Prize Global Accelerator

June 2023

**Project:** *Autonomous Recycling of Waste Fabric using Augmented Reality & Machine Learning*

As a Semi-Finalist in the global accelerator, enhanced our AR and ML-based project under the mentorship of international experts, refining scalability and operational efficiency for global implementation in the garments industry, and garnered significant acclaim and support

### Champion, Project Showcasing

April 2022

**Project:** *Automated Robotic Arm for Industrial Assistance Purpose*

*IEEE Robotics and Automation Society, ULAB Student Branch*

Led a team to develop a robotic arm enhancing efficiency and safety in industrial applications; awarded for excellence in design and implementation.

### 1<sup>st</sup> Runner-Up, Project Showcasing

April 2022

**Project:** *Auto-Balancing Two-Wheeler Vehicle with PID-Controller*

*IEEE Robotics and Automation Society, ULAB*

Recognized for developing a two-wheeler auto-balancing system with PID control, demonstrating advanced engineering skills.

*Project: Quality Assessment of Respiratory Sounds from Self-Assembled Digital Stethoscopes*

*IEEE Robotics and Automation Society, ULAB*

Awarded for presenting research on respiratory sound quality assessment using digital stethoscopes, highlighting diagnostic tool advancements.

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**PROJECTS****Enhancement of Catfish and Tilapia fisheries in Bangladesh using Microcontroller**

*Microprocessor and Embedded Systems Lab (EEE 311)*

Developed a microcontroller-based system to monitor and optimize water quality parameters like temperature, pH, and dissolved oxygen, significantly improving fish health and productivity in aquaculture.

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**Advanced Digital Signal Processing Techniques for Gender Differentiation in Voice Signals**

*Digital Signal Processing (EEE 3103)*

Applied advanced digital signal processing techniques, including spectral analysis and feature extraction, to enhance gender differentiation accuracy in voice signals, supporting applications in voice recognition.

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**Design and Performance of Conveyor Belt using PID for Industrial Applications**

*Control System-I (EEE 4103)*

Developed a PID-controlled conveyor belt system using a brushed permanent magnet DC motor, modeled in Simulink. Optimized system performance through precise PID tuning to achieve desired angular velocity and displacement, enhancing operational efficiency in industrial applications

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**PID Controller Optimization for Multilevel Inverters in Renewable Energy Applications**

*Power Electronics (EEE 433)*

Developed a PID controller-enhanced multilevel inverter configuration for optimizing energy conversion in renewable energy systems. The project focused on increasing the efficiency and stability of power output, crucial for the integration of renewable resources into the energy grid.

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**WORKSHOPS CONDUCTED****Expert and Mentor, Hult Prize OnCampus****Feb 2024**

*University of Liberal Arts Bangladesh (ULAB)*

Guided Hult Prize OnCampus participants by sharing strategic insights and practical solutions based on personal competition experiences. Focused on overcoming challenges, aligning projects with sustainable practices, and integrating startup ideas with the UN SDGs.

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**Facilitator, Raspberry Pi Applications****Dec 2023**

*ULAB Electronics & Robotics Club*

Led an interactive online workshop on Raspberry Pi applications, exploring its versatile uses in real-world projects and advanced computing solutions.

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**Facilitator, Overview on Arduino****Feb 2023**

*Department of Electrical and Electronic Engineering, ULAB*

Facilitated a workshop aimed at providing a comprehensive overview of Arduino applications, focusing on practical, hands-on implementations and innovative uses in technology projects.

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**TECHNICAL SKILLS****Programming and Development**

- *Languages:* Python, C-Programming
- *Machine Learning:* TensorFlow, Scikit-learn
- *Embedded Systems:* Arduino, ESP32, Raspberry Pi
- *Data Analysis & Visualization:* Jupyter Notebook, NumPy, Pandas

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**Design and Simulation**

- *Electrical Design:* MATLAB, PowerWorld Simulator, Proteus
- *CAD and 3D Modeling:* AutoCAD, Shapr3D
- *PCB Design:* KiCAD

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**Database and Web Development**

- *Database Management:* MySQL
  - *Web Technologies:* PHP MySQL, HTML5, CSS3
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## LANGUAGE PROFICIENCY

**Bangla**

Mother Tongue

**English**

*IELTS (International English Language Testing System)*

Score: 7.5 [Listening: 7.5; Reading: 7.5; Writing: 7.0; Speaking: 7.0]

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## Database and Web Development

- *Database Management:* MySQL
- *Web Technologies:* PHP MySQL, HTML5, CSS3

## EXTRA - CURRICULARS

- *Member, IEEE & RAS*
- *Guard, ULAB Basketball (Captain)*
- *Defender, EEE Football Team*
- *Guitar*

## REFERENCES

**Mirza Rasheduzzaman, PhD**

*Associate Professor*

Department of Electrical and Electronic Engineering, BRAC University

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**Abul Barkat Mollah Sayeed Ud Doulah, PhD**

*Associate Professor*

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